

HONGXIANG ZHANG

Lafayette, IN, USA ◇ [Homepage](#) ◇ [LinkedIn](#) ◇ [Google Scholar](#) ◇ hxxzhang@gmail.com ◇ (+1) 530-564-2580

RESEARCH INTERESTS

My research focuses on reliable and efficient AI systems, with an emphasis on LLM reasoning, multi-agent communication, hallucination mitigation, and AI for code. I am broadly interested in developing robust agentic workflows by combining alignment, decoding, and lightweight diagnostic methods across software engineering, security, and generative AI applications.

EDUCATION

Purdue University Ph.D in Computer Science; Advised by Prof. Tianyi Zhang	August 2025 - Present
University of California, Davis M.Sc. in Computer Science	Sep 2022 - June 2025
Australian National University B.Eng. (Honours) in Computer Science	Jul 2020 - Sep 2022
Shandong University B.Eng. in Computer Science	Sep 2018 - Jun 2022

ACADEMIC EXPERIENCE

Enhancing Multi-Agent Communication through Attention Steering with Context Relevance

Hongxiang Zhang, Yuan Tian, Tianyi Zhang

[\[Under review\]](#) | Multi-agent systems; Attention Alignment; Context Management

- Proposed a training-free context management method that steers each agent's attention toward relevant conversation history using temporal and spatial decay.
- Improved multi-agent performance across five benchmarks by up to 7.64 absolute points while remaining robust as the number of agents and interaction rounds increases.

FLARE: Fine-Grained Diagnostic Feedback for LLM Code Refinement

Yinsheng Yao, Hongxiang Zhang, Weixi Tong, Tianyi Zhang

[\[Under review\]](#) | Code Refinement; Fault Localization; Test-time Scaling

- Proposed an iterative code-refinement framework centered on a lightweight fault localizer that predicts line-level suspiciousness for targeted LLM refinement.
- Improved LiveCodeBench and BigCodeBench performance across five LLMs, with up to 7.42% gain without candidate search and 8.50% average gain with top- k search.

Attention-Aligned Reasoning for Large Language Models

Hongxiang Zhang, Yuan Tian, Tianyi Zhang

[\[arXiv\]](#)

[\[Under review\]](#) | LLM Reasoning; Attention Alignment; Prompt Engineering

- Proposed a reasoning-time alignment method that uses step-by-step attention alignment to keep LLMs focused on the prompt and intermediate goals.
- Outperformed existing prompting-based approaches and RL-trained SLMs.

Active Layer-Contrastive Decoding Reduces Hallucination in Large Language Model Generation

Hongxiang Zhang, Hao Chen, Muhao Chen, Tianyi Zhang

[\[arXiv\]](#)

[\[EMNLP 2025 Main\]](#) | Hallucination Mitigation; Decoding; Reinforcement Learning

- Proposed an RL-guided decoding strategy that actively contrasts model layers to suppress hallucinations.
- Achieved state-of-the-art hallucination reduction across five generation benchmarks.

LLAMAFUZZ: Large Language Model Enhanced Greybox Fuzzing

Hongxiang Zhang, Yuyang Rong, Yifeng He, Hao Chen

[arXiv]

[AST 2026] | LLM for Systems; Greybox Fuzzing; Software Testing

- Built LLAMAFUZZ, an LLM-enhanced greybox fuzzer that learns structured input formats and mutation strategies for binary and text targets.
- Improved real-world coverage and bug finding, identifying 47 unique bugs and increasing branch coverage by 27.19% over AFL++.

SteerDiff: Steering towards Safe Text-to-Image Diffusion Models

Hongxiang Zhang, Yifeng He, Hao Chen

[arXiv]

Safe Generative AI; Prompt Steering; Diffusion Models

- Developed **SteerDiff**, a lightweight adapter that steers prompt embeddings away from unsafe concepts before text-to-image diffusion.
- Preserved generation quality while improving robustness against red-teaming and supporting concept forgetting tasks.

TEACHING EXPERIENCE

ECS 153 Computer Security - UC Davis	2024 Spring
ECS 036A Programming & Problem Solving - UC Davis	2024 Winter
ECS 036C Data Structures, Algorithms, & Programming - UC Davis	2023 Fall
ECS 140A Programming Language - UC Davis	2022 Winter
ENGN 4528 Computer Vision - Australian National University	2021

INDUSTRY EXPERIENCE

Software and Operation Engineer Intern Volkswagen, Beijing, China	Feb 2022 - Jul 2022
· Achieved data masking and auto-populated to JIRA log in Python. · Intuitively analyzed and provided services and finance data to CIO with Tableau, helping the management level adjust the company service strategy agilely.	
Quality Assurance Intern Didi Global, Beijing, China	Dec 2020 - Feb 2021
· Developed distributed Java invoice services to enable the interaction between server and end-user devices that served more than 1 million users per day. · Achieved unit testing automation by using JUnit and the test case in Redis, reduced over 15% of the testing engineer's workload.	

HONORS AND AWARDS

Microsoft Certified: Azure Fundamentals	2022
Shandong University Scholarship	2018- 2019
Shandong University Research and Innovation Specialty Scholarship	2018- 2019
Lan Qiao Collegiate Cup Provincial Third Award	2019
Shandong University Scholarship	2020- 2021
Excellent Course Representative of ENGN4528/6528	2021